

Evolution Rather Than Degradation: Structure-Guided Elastic Consensus Learning for Multimodal Knowledge Graph Completion

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Abstract—Recent efforts in multimodal knowledge graph completion (MKGC) have demonstrated impressive progress empowered by contrastive learning. However, a non-common sense observation shows that the simple pursuit of cross-modal consensus might degenerate MKGC performance unexpectedly, owing to the integration of untrustworthy auxiliary modalities. This behavior is herein referred to as *modality degradation*, which inadvertently diminishes the synergy between contrastive learning and MKGC. In this paper, we introduce a Structure-Guided Elastic Consensus Learning (SECL) framework. It captures cross-modal consensus from trustworthy multimodal entity representations for MKGC. Specifically, to reduce the semantic bias observed in CLIP-derived multimodal representations, a structure-guided multimodal representation harmonization method is introduced in SECL. With the structural context of different entities being a coordinator, this method selectively suppresses irrelevant features in CLIP-derived multimodal representations, thereby promoting their inter-modal semantic consistency. To supervise this harmonization process, an elastic contrastive learning method is presented. It alleviates the under-exploration of cross-modal consensus by steering the model optimization to prioritize aligning multimodal representations for hard entities. Extensive experiments on benchmark datasets show that SECL exhibits *modality evolution* instead of *degradation* and yields substantial improvements over state-of-the-art methods.

Index Terms—Multimodal knowledge graph, knowledge graph completion, contrastive learning, modality alignment.

I. INTRODUCTION

KNOWLEDGE graphs (KGs) have demonstrated remarkable vitality across a wide range of real-world scenarios like healthcare [1], e-commerce [2], and finance [3]. However, KGs often suffer from incompleteness because of sub-optimal

construction pipelines [4], which makes considerable amounts of knowledge still undiscovered. This significantly weakens its practical applicability in real-world scenarios. In this context, knowledge graph completion (KGC) has emerged as a promising direction to alleviate the problem by inferring missing facts and expanding the knowledge base.

Recent years have witnessed striking progress in KGC [6], [7], [8], [9], [10]. Conventional methods mainly learn unimodal entity representations by leveraging the structural information contained in triples. To go beyond the structural information, multimodal knowledge graph completion (MKGC) approaches have been striving to use auxiliary (e.g., visual and textual) modalities to further enhance entity representations. In contrast to unimodal approaches, MKGC has attracted increasing attention for better capturing the complexities inherent in real-world scenarios. In order to appropriately use these auxiliary modalities, dominant methods [11], [12], [13] adopt the modality-fusion paradigm (MFP), which allows for capturing complementary information across modalities through integrating multimodal features into unified entity representations. However, *modality-specific information is readily sacrificed during propagation*. Another line of work [5] adopts the modality-ensemble paradigm (MEP). It retains modality-specific information through modeling each modality independently. Unfortunately, *the inter-modal interdependency is under-explored*, which prevents models from grasping entities from a holistic perspective.

To tackle these issues, previous studies attempt to introduce the inter-modal interdependency into MEP. Beyond modality-independent representations, a unified representation that fuses distinct modalities [14], [15] is derived to assist in uncovering missing triples within multimodal knowledge graphs (MKGs). Another perspective is to leverage the inter-modal interdependency by casting different modalities as reciprocal supervisory signals, which has been demonstrated more effective by recent studies [16], [17]. They promote the alignment of cross-modal representations through contrastive learning [18], [19], [20], thereby exploiting cross-modal consensus of entities to enhance entity representations for MKGC.

Despite yielding significant progress in previous methods, a non-common sense observation reveals that the simple pursuit of modality consensus might degenerate MKGC performance, termed *modality degradation* here, in which compulsory inter-modal alignment for all modality pairs would inevitably lead to

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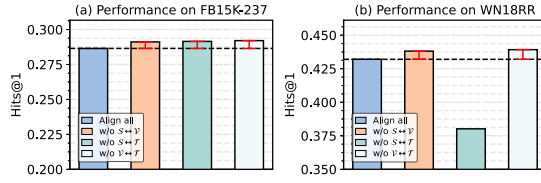


Fig. 1. Illustration of modality degradation. The red lines denote performance gaps. In the MKG that simultaneously includes structural (S), visual (V), and textual (T) information, the simple pursuit of cross-modal alignment between all modalities results in suboptimal performance. Experiments are conducted based on [5], with the InfoNCE objective as an optimization constraint.

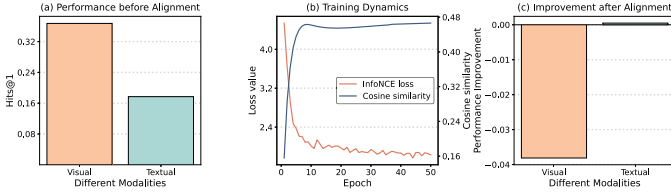


Fig. 2. Effect of untrustworthy modalities. (a) Performance of two modalities on WN18RR before alignment. (b) Training dynamics of contrastive learning. (c) Performance improvement of two modalities after alignment.

suboptimal performance. In order to quantitatively measure the degree of degradation, some targeted preliminary experiments are presented in Fig. 1. As we can see, simply enforcing alignment across all modality pairs consistently induces suboptimal MKGC performance, which weakens the synergy between CL and MKGC. This raises a question of why enforcing alignment across all modalities could be detrimental. *A critical factor lies in the integration of untrustworthy modalities.* In particular, we illustrate it in Fig. 2 by taking the visual and textual modalities as an example. Clearly, there exists an inherent trustworthiness gap between these two modalities, which is manifested in their respective MKGC performance. When a contrastive constraint is introduced between them, the better-performing modality is required to align with the poorer-performing one. This directly contaminates the trustworthy feature space with noise from the untrustworthy modality. Generalizing this case to all pairwise modalities in MKGs, such noise will diffuse across the feature spaces of other modalities. As a result, these multimodal entity representations become mediocre, thus resulting in suboptimal performance. Recognizing this as a potential cause of modality degradation, *the challenge lies in how to purify untrustworthy multimodal representations into trustworthy ones.*

To address this problem, this paper introduces a **Structure-Guided Elastic Consensus Learning (SECL)** framework¹. It promotes MKGC by achieving *modality evolution rather than degradation*, which indicates that SECL benefits from, instead of being impaired by, different modality pairs. As illustrated in Fig. 3, different from the general contrastive learning paradigm [21], [22], [23], [24] that focuses on visual and textual data, the structural information within MKGs is incorporated into our framework. Although structural information has been explored in existing MKGC methods [5], [15], they merely treat

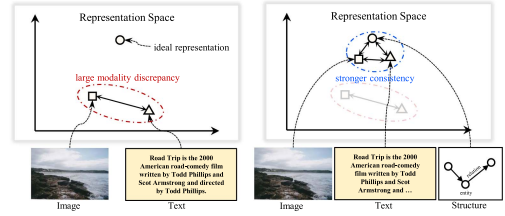


Fig. 3. Illustration of (left) general contrastive learning and (right) ours.

it as an additional modality to be integrated with the visual and textual features. Going beyond them, this study, **to the best of our knowledge**, is the first to cast structural information as a *coordinator* that recalibrates visual and textual representations into trustworthy ones. Thanks to the stable relational dependencies within KGs, the semantic bias within visual and textual representations can be alleviated to achieve stronger cross-modal consistency.

Specifically, to mitigate the inherent semantic bias observed in the CLIP-derived multimodal representations, this paper introduces a structure-guided multimodal representation harmonization (SMRH) method motivated by Canonical Correlation Analysis (CCA) [25]. With the structural context of different entities being a coordinator, this method selectively suppresses semantically irrelevant features in visual and textual representations *at an entity level*. Moreover, to achieve the goal of enhancing cross-modal consistency by suppressing semantically irrelevant features, we introduce an elastic contrastive learning (ECL) method. Different from the conventional paradigm [20] that treats all entities equally, our ECL prioritizes speeding up learning on hard entities and slowing down that on easy ones based on their cross-modal consistency. In this way, it lessens the under-exploration of cross-modal consensus, thus securing the semantic understanding of entities. Viewing from an overall perspective, the proposed SMRH and ECL are *complementary*: The former can furnish the latter with trustworthy multimodal representations and, in turn, the latter provides powerful supervisory signals for the former. The seamless integration of them enables our framework to counteract modality degradation by learning trustworthy multimodal representations. In a nutshell, the main contributions of this paper are highlighted as follows:

- This work identifies the dilemma of modality degradation, providing a quantification of its extent via empirical analysis. It shows the adverse impact of modality degradation on the synergy between MKGC and CL.
- To model trustworthy multimodal representations, SMRH casts the structural context of different entities as a coordinator and selectively suppresses semantically irrelevant features in CLIP-derived representations.
- To provide powerful supervisory signals for SMRH, ECL is proposed to prioritize aligning cross-modal representations for hard entities. In this way, it mitigates the under-exploration of cross-modal consensus.
- Extensive experiments show that SECL exhibits modality evolution rather than degradation. Across five benchmark datasets, it consistently demonstrates substantial improvements over state-of-the-art methods.

¹Code is available at <https://github.com/ymlu-felix/SECL>.

The rest of this paper is organized as follows: A brief review of related work is introduced in Section II. Section III shows necessary preliminaries related to this work. Section IV details the proposed framework. Experimental results and analysis are presented in Section V. Section VI concludes this paper.

II. RELATED WORK

A. Unimodal Knowledge Graph Completion

Unimodal methods primarily discover missing relations using structural information from triples. Among these methods, translation-based methods are the most representative, such as TransE [26]. They naturally cast the relation from a head entity to a tail entity as a translation in the latent space. In order to effectively capture different relational patterns, ComplEx [27] maps the embedding of entities and relations into the complex space, and its theoretical analysis demonstrates that complex space tensor operations naturally fit various relational patterns in entity-relation interactions. Tucker [28] exploits the Tucker tensor decomposition to reverse the entity-relation interaction. Motivated by the application of convolution in computer vision [29], ConvE [30] and LTE-ConvE [31] leverage convolutional neural networks to capture the interaction between the entity and relation. NFE [32] proposes a group theory-based method that embeds entities/relations as symmetric group elements and incorporates uncertainty via normalizing flows. CircularE [33] leverages complex-space circular correlation to enhance entity-relation interactions without expanding space dimension. Another interesting perspective is to adopt graph neural networks for KGC [34]. SE-GNN [35] identifies three levels of semantic evidence that support embedding extrapolation, and explicitly models them to improve extrapolation ability. DRGI [6] deals with the graph disconnectedness arising from incomplete facts by capturing the semantic and complete structural information of KGs. To model the contextual information of nodes in KGs, TGformer [9] simultaneously fuses triple-level and graph-level structural information. Despite being effective, *these methods overlook multimodal information associated with entities*, which is also important for grasping semantics [36], [37], [38].

B. Multimodal Knowledge Graph Completion

To overcome the inherent bottleneck of unimodal methods, MKGC has attracted increasing attention. Specifically, to use visual information, IKRL [39] integrates visual representations with triple facts for knowledge representation learning. RSME [40] adaptively integrates visual context into knowledge graph embeddings by modulating image influence based on relation-specific semantics. MKGformer [11] introduces a hybrid transformer with multi-level fusion to discover missing relations in MKGs. Based on this, SGMPT [41] explicitly introduces graph structure into transformers for knowledge graph reasoning. To obtain reliable entity representations, LAFA [12] dynamically fuses visual and structural embeddings based on importance. To capture the fine-grained semantic details in descriptions and images associated with entities, MyGO [13]

tokenizes multi-modal entity information into discrete fine-grained tokens and fuses them using an encoder for MKGC. *These methods adopt MFP, readily sacrificing modality-specific information* that is indispensable for capturing the characteristics of entities.

While MoSE [5] proposes MEP to retain modality-specific signals, *it ignores the inter-modal interdependency*. To tackle this issue, MoCi [14] achieves inter-entity multimodal fusion via multilinear transformation, thus learning a joint representation to capture inter-modal correlations. MoMoK [15] exploits relation-guided experts and adaptive fusion to derive additional unified representations, with mutual information minimization for expert disentanglement. Motivated by the success of CL in diverse scenarios [18], [19], CL has been increasingly adopted as a robust framework to enhance representation learning for MKGC. IMF [16] maintains modality-specific representations while considering inter-modal interactions via bilinear pooling and contrastive learning. HKA [17] aligns cross-modal global semantics via macro-alignment and achieves fine-grained consistency through inter- and intra-modal supervision in micro-alignment. Compared to [14], [15], these methods yield superior MKGC performance by mining the cross-modal consensus of entities. Yet, modality degradation induced by untrustworthy multimodal representations remains under-explored.

III. PRELIMINARY

Notations: A multimodal knowledge graph is represented as $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{F})$, where $\mathcal{E}$ and $\mathcal{R}$ denote the set of entities and relations, respectively. $\mathcal{F} = \{(h, r, t) | h, t \in \mathcal{E}, r \in \mathcal{R}\}$ is the set of factual triples, where r represents the relation between head entity h and tail entity t . For any entity $e_i \in \mathcal{E}$ in $\mathcal{G}$, it is simultaneously associated with a textual description $\mathbf{e}_i^T$ and M images $\mathbf{e}_i^V = \{\mathbf{e}_i^V(1), \dots, \mathbf{e}_i^V(M)\}$, where $\mathcal{T}$ and $\mathcal{V}$ represent textual and visual modalities, respectively. Collectively, for $q \in \{\mathcal{T}, \mathcal{V}\}$, let $\mathbf{E}^q = [\mathbf{e}_1^q, \dots, \mathbf{e}_n^q]$ represent the whole textual or visual data. Beyond that, each entity is equipped with implicit structural information $\mathcal{S}$ that reflects its relational dependency in $\mathcal{G}$. For ease of description, we could define the modality set as $\mathcal{M} = \{\mathcal{S}, \mathcal{V}, \mathcal{T}\}$ throughout this paper.

Problem Formulation: Given the MKG $\mathcal{G}$, MKGC targets to predict a plausibility score to the given triple (h, r, t) via learning a mapping function $\mathcal{K}_\Theta(h, r, t) : \mathcal{E} \times \mathcal{R} \times \mathcal{E} \rightarrow \mathbb{R}$, where Θ represents parameters. Notably, a higher score suggests a more plausible triple. In the inference stage, trained MKGC models predict missing entities for the tail-missing triple $(h, r, ?)$ and head-missing triple $(?, r, t)$. Taking the former as an example. Given the head entity and relation, MKGC models regard each entity $e_i \in \mathcal{E}$ as a candidate and compute its plausibility score $\mathcal{K}_\Theta(h, r, e_i)$. According to the scores of all candidates, MKGC models are evaluated through rank-based metrics.

Empirical Definition of Modality Degradation and Evolution: Given the set of modality pairs $\mathcal{P}$, let $A_{\mathcal{P}}$ be the model performance when all modality pairs in $\mathcal{P}$ are employed, and let $A_{\mathcal{P} \setminus (u, v)}$ denote the performance after removing a specific modality pair (u, v) , where $u, v \in \mathcal{M}$. In this context, modality degradation happens if $\exists (u, v) \in \mathcal{P}$ such that $A_{\mathcal{P}} < A_{\mathcal{P} \setminus (u, v)}$.

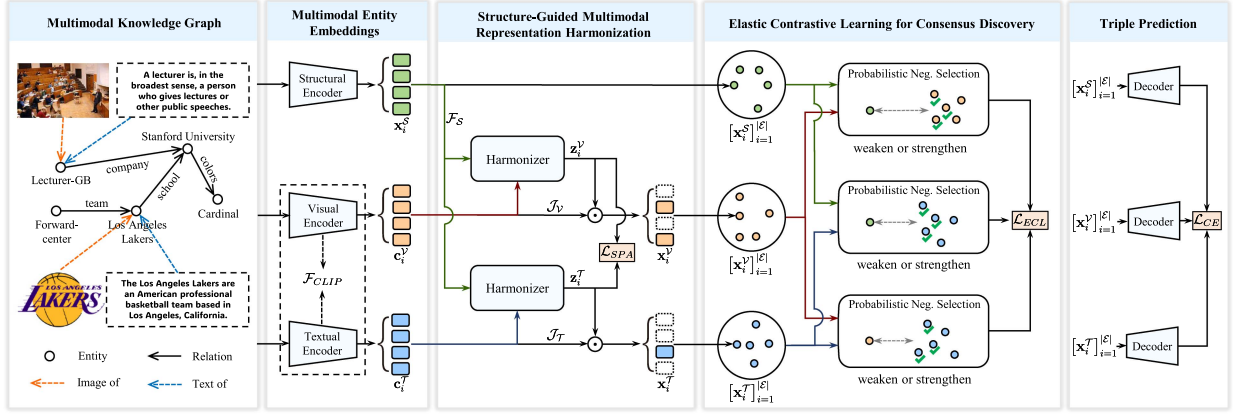


Fig. 4. Overview of our SECL framework. With the assistance of the structural context $\mathbf{x}_i^S$, the semantically irrelevant features within CLIP-derived embeddings are suppressed via SMRH, deriving the debiased visual representation $\mathbf{x}_i^V$ and textual representation $\mathbf{x}_i^T$. Based on these multimodal representations, ECL prioritizes the alignment for hard entities to prevent their under-discovery of cross-modal consensus. Finally, a decoder is used to predict plausibility scores.

In contrast, modality evolution occurs if $A_p > A_{p \setminus (u,v)}$ when $\forall (u, v) \in \mathcal{P}$.

Objective: Given that modality degradation is a specific manifestation of an untrustworthy alignment procedure attributed to semantic biases, the core objective of this study is to establish a trustworthy cross-modal consensus learning framework. By learning multimodal representations with essential semantics, the interference of untrustworthy alignment is mitigated, thus steering the model from degradation toward evolution.

IV. METHODOLOGY

A. Overview of the SECL Framework

This paper presents SECL to tackle the dilemma of modality degradation induced by untrustworthy cross-modal alignment. The overall framework is depicted in Fig. 4, mainly consisting of the following components:

- **Multimodal Entity Embeddings** comprise structural, visual, and textual embeddings, which capture the semantic information of entities from complementary views.
- **Structure-Guided Multimodal Representation Harmonization** formulates the structural context of entities as a coordinator, suppressing semantically irrelevant features in CLIP-derived representations to model reliable multimodal entity representations for consensus discovery.
- **Elastic Contrastive Learning for Consensus Discovery** is to steer the optimization procedure to prioritize aligning multimodal representations for hard entities, such that reducing their under-exploration of cross-modal consensus while providing supervisory signals for the representation harmonization process.

B. Multimodal Entity Embeddings

Visual and Textual Embeddings: CLIP [42] has encapsulated a myriad of cross-modal knowledge via pre-training on large-scale image-caption pairs, showing a highly enriched semantic

space [43]. For any given image-text pair, the shared semantic information could be initially mined by CLIP, offering a well-initialized representation space for consensus discovery across modalities. Hence, it is highly appropriate to leverage CLIP to encode both images and descriptions of different entities in $\mathcal{G}$. Formally, let $\mathbf{C}^V = [\mathbf{c}_1^V, \dots, \mathbf{c}_{|\mathcal{E}|}^V]$ and $\mathbf{C}^T = [\mathbf{c}_1^T, \dots, \mathbf{c}_{|\mathcal{E}|}^T]$ denote the visual and textual embeddings encoded by the CLIP encoder $\mathcal{F}_{CLIP}$, respectively. Notably, each entity is described by M images, which are compacted into a joint representation using learnable weights:

$$\mathbf{c}_i^V = \sum_{j=1}^M \frac{\exp(s_{i,j})}{\sum_{k=1}^M \exp(s_{i,k})} \mathcal{F}_{CLIP}(\mathbf{e}_i^V(j)), \quad (1)$$

where $s_{i,j}$ is a trainable weight.

Structural Embeddings: Beyond visual and textual information within MKGs, structural information is also indispensable for comprehending the semantics of entities because it encodes comprehensive relational context among entities. To explicitly represent structural information, a learnable structural encoder is introduced to map each entity in the MKG into a dedicated structural embedding. Formally, the structural embedding matrix is written as $\mathbf{X}^S = [\mathbf{x}_1^S, \dots, \mathbf{x}_{|\mathcal{E}|}^S] \in \mathbb{R}^{d_s \times |\mathcal{E}|}$. Essentially, under the supervision of triple decoders (e.g., ComplEx [27]), these structural embeddings are optimally established to model the topological structures of MKGs, serving as a reliable prior to guide multimodal representation learning.

C. Structure-Guided Multimodal Representation Harmonization

In real-world scenarios, a large amount of entity-irrelevant information is introduced into MKGs owing to the suboptimal processes in collecting multimedia data. Even though extensive cross-modal knowledge has been encapsulated in CLIP [42], it still exhibits notable semantic bias when faced with such challenging multimodal data. As shown in Fig. 5(a), CLIP-derived multimodal representations consistently exhibit low correlation on different MKGC datasets, showing unsatisfactory semantic

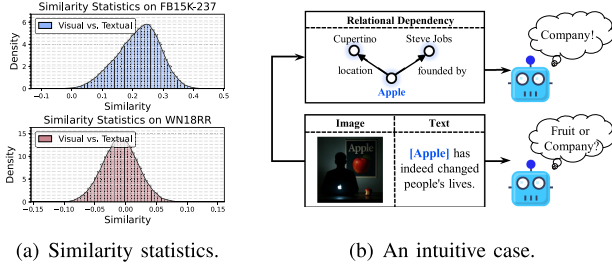


Fig. 5. (a) Statistics of the cosine similarity between CLIP-derived visual and textual embeddings. (b) An intuitive case showing how the structural context of an entity helps mitigate semantic ambiguity.

consistency between cross-modal representations. In this case, the simple pursuit of alignment across these native multimodal representations will gradually render powerful representations mediocre due to the intrusion of entity-irrelevant information, thereby inducing modality degradation. Hence, it becomes critical to obtain trustworthy multimodal representations through reducing semantic biases in them.

Compared with visual and textual data that usually include irrelevant background information, structural information can encapsulate more reliable relational dependencies in MKGs. Indirectly, it provides an invariant semantic structure that helps reduce semantic ambiguity in multimodal data. As illustrated in Fig. 5(b), it is not trivial to determine whether *Apple* refers to a *fruit* or a *company* based on the image and text, whereas incorporating the knowledge graph can effectively resolve such semantic ambiguity. Hence, it becomes natural to debias visual and textual representations using structural information.

In practice, a straightforward way to achieve this is through Canonical Correlation Analysis (CCA) [25]. It seeks to simultaneously learn nonlinear transformations for the two different views, such that the resulting representations can be maximally correlated in the latent space. Mathematically, when employing structural information to coordinate the modality $q \in \{\mathcal{V}, \mathcal{T}\}$ for debiasing, it can be formulated as the following CCA-style optimization problem:

$$\begin{aligned} \min \quad & \|\Phi_S(\mathbf{X}^S) - \Phi_q(\mathbf{E}^q)\|_F^2, \\ \text{s.t.} \quad & \Phi_S(\mathbf{X}^S) (\Phi_S(\mathbf{X}^S))^\top = \Phi_q(\mathbf{E}^q) (\Phi_q(\mathbf{E}^q))^\top = \mathbf{I}, \end{aligned} \quad (2)$$

where Φ_S and Φ_q denote mapping functions, and $\|\cdot\|_F$ is the Frobenius norm. Given the optimal Φ_S and Φ_q by solving this optimization problem, the shared representation $\mathbf{X}_{CCA}^q$ can be obtained as:

$$\mathbf{X}_{CCA}^q = \frac{1}{2} (\Phi_S(\mathbf{X}^S) + \Phi_q(\mathbf{E}^q)), \quad (3)$$

where the semantic bias in modality q is mitigated through the structural information of MKGs. However, it can be observed that the mapping Φ_q in modality q is shared across all entities, which implicitly assumes that similar modality-specific biases are exhibited by all entities. Such an assumption overlooks the diversity of entity representations, readily leading to undercorrection or overcorrection for certain entities.

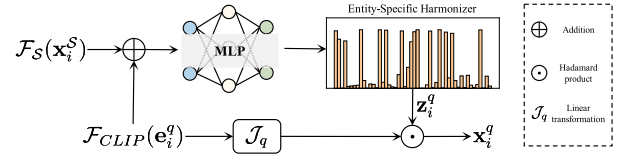


Fig. 6. Illustration of our SMRH. The entity-specific harmonizer $\mathbf{z}_i^q$ is induced by structural context, selectively suppressing features in the visual and textual representation and deriving debiased multimodal representations $\mathbf{x}_i^q$.

To address this problem, we propose a structure-guided multimodal representation harmonization (abbreviated by SMRH) method. Unlike CCA that is restricted to coarse-grained debiasing via the global nonlinear transformation, SMRH enables *fine-grained* debiasing by learning entity-specific harmonizers, which explicitly suppress those semantically irrelevant features within visual and textual representations. As shown in Fig. 6, for $q \in \{\mathcal{V}, \mathcal{T}\}$, the debiased representation $\mathbf{x}_i^q$ is defined as:

$$\mathbf{x}_i^q = \mathbf{z}_i^q \odot \mathcal{J}_q(\mathbf{c}_i^q), \quad (4)$$

where $\odot$ represents the Hadamard product and $\mathcal{J}_q$ indicates a linear transformation that projects the input into the representation space of $\mathbf{z}_i^q \in \mathbb{R}^{d_s}$. The entity-specific harmonizer $\mathbf{z}_i^q$ is induced by the structural context as:

$$\mathbf{z}_i^q = \text{MLP}(\mathcal{F}_{CLIP}(\mathbf{e}_i^q) + \mathcal{F}_S(\mathbf{x}_i^S)), \quad (5)$$

where $\hat{\mathbf{z}}_i^q = \mathcal{F}_{CLIP}(\mathbf{e}_i^q) + \mathcal{F}_S(\mathbf{x}_i^S)$ denotes CCA-style shared representation as in (3) and $\mathcal{F}_S(\cdot)$ denotes a linear projection. Moreover, to make the harmonizer $\mathbf{z}_i^q$ more semantically sensitive for debiasing, a sparsity regularization term $\mathcal{L}_{SPA}$ is introduced, which is formulated as:

$$\mathcal{L}_{SPA} = \sum_{q \in \{\mathcal{V}, \mathcal{T}\}} \sum_i \|\mathbf{z}_i^q\|_1, \quad (6)$$

where $\|\cdot\|_1$ denotes the L_1 norm. In summary, the harmonizer can attenuate partial features in multimodal representations by estimating lower relevance for them with the help of the structural context of entities. However, at the current stage, there is no explicit regularization that encourages the proposed SMRH to suppress those semantically irrelevant features. In contrast, this might diminish relevant ones and thereby compromise the semantic integrity of multimodal representations.

D. Elastic Contrastive Learning for Consensus Discovery

To tackle the problem from Section IV-C, an explicit regularization is incorporated to supervise the harmonization procedure. Inspired by the success of contrastive learning (CL) [14], [17], it is introduced to discover cross-modal consensus of entities. By encouraging the alignment of cross-modal representations, it provides supervisory signals that instruct SMRH to suppress semantically irrelevant features. In this way, trustworthy multimodal representations are optimally derived through debiasing, such that modality degradation can be mitigated.

Specifically, representations of the same entity are regarded as positive pairs, whereas those of different entities are cast as negative pairs. Because the unidirectional contrastive objective

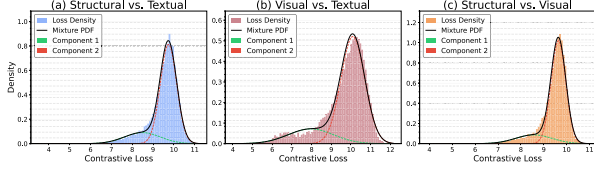


Fig. 7. Analysis of the InfoNCE losses across all entities in the FB15K-237 dataset. The black solid line and dashed lines represent the estimated Mixture Probability Density Function (PDF) and two underlying Gaussian components disentangled by GMM, respectively.

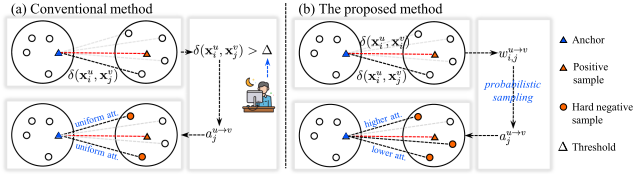


Fig. 8. (a) Conventional methods depend on a predefined threshold, restricting their flexibility to diverse data distributions. (b) Our method is threshold-free, allowing harder negatives to contribute more to consensus discovery.

tends to yield biased optimization, a symmetric variant is used to facilitate balanced alignment across modalities:

$$\mathcal{L}_{CL}^{u,v}(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N [\mathcal{L}(\mathbf{x}_i^{u \rightarrow v}) + \mathcal{L}(\mathbf{x}_i^{v \rightarrow u})],$$

$$\mathcal{L}(\mathbf{x}_i^{u \rightarrow v}) = -\log \frac{\exp(\delta(\mathbf{x}_i^u, \mathbf{x}_j^v)/\tau)}{\sum_j \exp(\delta(\mathbf{x}_i^u, \mathbf{x}_j^v)/\tau)}, \quad (7)$$

where $u, v \in \mathcal{M}$ with $u \neq v$, δ stands for cosine similarity, and τ is the temperature. It is not hard to observe that each entity is assigned an identical weight $1/N$. However, as shown in Fig. 7, a large portion of entities maintain higher contrastive losses across distinct modality pairs, demonstrating that mining their cross-modal consensus poses a greater challenge than it does for entities with lower losses (respectively called hard and easy entities). In this context, because of the *memorization effect* of Dense Neural Networks,² traditional paradigm could result in the under-discovery of cross-modal consensus for hard entities. To this end, elastic contrastive learning (ECL) is introduced. It could adaptively scale the weights of entities according to their cross-modal consistency and steer the model optimization to prioritize aligning multimodal representations for hard entities. This offers SMRH with more powerful supervisory signals to model trustworthy multimodal representations, thus mitigating modality degradation.

Probabilistic Negative Selection: Contrastive learning mainly benefits from hard negatives [44]. Most hard negative mining methods [45], [46] depend on predefined thresholds, as illustrated in Fig. 8(a). This restricts their flexibility to diverse data distributions. To this end, we propose a threshold-free selection method, as depicted in Fig. 8(b). The key insight behind this is that negatives whose similarity to the anchor is comparable to

²The memorization effect characterizes the learning dynamics of DNNs, whereby they tend to fit easy samples first and gradually fit harder ones.

or exceeds that of the positive pair are more likely to qualify as hard negatives. In practice, for i -th entity, a form of relative similarity can be formulated as:

$$w_{i,j}^{u \rightarrow v} = \min \left(1, \frac{\delta(\mathbf{x}_i^u, \mathbf{x}_j^v)}{\delta(\mathbf{x}_i^u, \mathbf{x}_i^v)} \right), \quad (8)$$

where $w_{i,j}^{u \rightarrow v}$ quantitatively represents how likely a candidate negative is to be a hard negative. Diverging from conventional hard negative mining methods, our method offers a continuous signal that can be used for *probabilistic selection*. Specifically, instead of deterministically selecting those negatives exceeding a threshold, we regard the sampling decision of each negative as a stochastic event, parameterized by the relative similarity. Based on this thinking, we have the following claim:

Claim 1: Given the relative similarity $w_{i,j}^{u \rightarrow v}$ between a candidate negative $\mathbf{x}_j^v$ and the anchor $\mathbf{x}_i^u$, the negative sampling process can be modeled as a series of independent Bernoulli trials, where each candidate negative is sampled with success probability $w_{i,j}^{u \rightarrow v}$ independently of others.

This implies that each candidate negative is sampled independently according to a *Bernoulli* sampling:

$$a_j^{u \rightarrow v} \sim \text{Bernoulli}(w_{i,j}^{u \rightarrow v}), \quad (9)$$

where $a_j^{u \rightarrow v} \in \{0, 1\}$ represents whether the j -th candidate in modality $v \in \mathcal{M}$ is selected as a hard negative sample.

Consistency-Preserving Entity Division: Driven by the memorization effect of DNNs, the difference of loss distribution is leveraged to distinguish the learning difficulty of entities in $\mathcal{G}$. To be specific, the per-entity loss can be formulated as $\{\ell_i\}_{i=1}^{|E|}$ for $u, v \in \mathcal{M}$ with $u \neq v$, which is calculated by the native contrastive learning objective that integrates hard negatives in (9). In order to select hard entities, these losses are fed into a two-component Gaussian Mixture Model (GMM) formulated as $p(\ell|\theta) = \sum_{k=1}^K \beta_k \phi(\ell|k)$, where β_k and $\phi(\ell|k)$ are mixture coefficient and the probability density of the k -th component, respectively. The easy entities are assigned the component with a lower mean, and the hard entities are given another one. In this process, we use the Expectation-Maximization algorithm [47] to optimize GMM and calculate the posterior probability $p(k|\ell_i) = p(k)p(\ell_i|k)/p(\ell_i)$ as the probability of i -th entity of being a hard entity, where k is the Gaussian component with the higher mean. Essentially, $p(k|\ell_i)$ represents the confidence level. Following the canonical decision boundary, we define a critical probability ϑ_p to divide entities. Formally, hard entities are classified as $\mathcal{Q} = \{e_i | p(k|\ell_i) > \vartheta_p\}$ and the rest are easy.

Elastic Contrastive Learning Objective: To stably update the importance of entities during training, we follow the boosting strategy [48] and introduce a recursive way, as depicted in Fig. 9. Specifically, the learning importance λ_i of the entity e_i can be formulated as:

$$\lambda_i^{(T+1)} = \begin{cases} \lambda_i^{(T)} \exp(\alpha_T) p(k|\ell_i), & \text{if } e_i \in \mathcal{Q} \\ \lambda_i^{(T)} \exp(-\alpha_T) (1 - p(k|\ell_i)), & \text{otherwise} \end{cases} \quad (10)$$

where T is the training epoch, $\alpha_T = \frac{1}{2}[\ln(1 - \Lambda_T) - \ln(\Lambda_T)]$ can be calculated according to $\Lambda_T = \sum b_i \lambda_i^{(T)}$, where $b_i = 1$ when $e_i \in \mathcal{Q}$ and $b_i = 0$ otherwise. Once all the importance

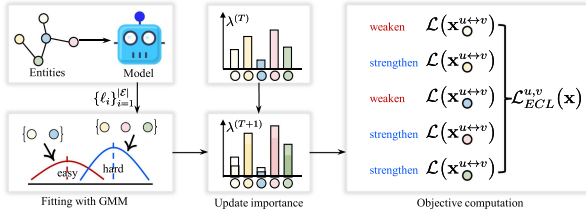


Fig. 9. Illustration of the elastic contrastive learning objective. The per-sample loss is fitted with GMM to select hard entities at the $(T+1)$ -th epoch, which together with $\lambda^{(T)}$ is the basis to generate the learning importance $\lambda^{(T+1)}$.

weights are updated, the normalization is adopted to conform to a distribution. Since there is no prior information to reliably distinguish hard entities in advance, $\lambda_i^{(0)}$ is initialized to $1/|\mathcal{E}|$, such that providing an unbiased starting point for each entity. At this stage, a difficulty-aware contrastive learning objective can be formulated as:

$$\mathcal{L}_{ECL}^{u,v}(\mathbf{x}) = \sum_{i=1}^N \lambda_i [\mathcal{L}(\mathbf{x}_i^{u \rightarrow v}) + \mathcal{L}(\mathbf{x}_i^{v \rightarrow u})],$$

$$\mathcal{L}(\mathbf{x}_i^{u \rightarrow v}) = -\log \frac{\exp(\delta(\mathbf{x}_i^u, \mathbf{x}_i^v)/\tau)}{\sum_{j \in \mathcal{N}_i^{u \rightarrow v}} w_{i,j}^{u \rightarrow v} \exp(\delta(\mathbf{x}_i^u, \mathbf{x}_j^v)/\tau)}, \quad (11)$$

where $\mathcal{N}_i^{u \rightarrow v}$ represents the set of selected negative samples. Finally, all pairwise modalities in $\mathcal{M}$ are contrasted to capture cross-modal consensus. Moreover, since learnable weighting is leveraged to adaptively integrate multiple visual embeddings, a contrastive constraint is further imposed between $\mathbf{c}_i^v$ and $\mathbf{c}_i^{\tau}$ to encourage the selection of semantically relevant images. The overall ECL objective is formulated as:

$$\mathcal{L}_{ECL} = \sum_{u,v \in \mathcal{M}} \mathcal{L}_{ECL}^{u,v}(\mathbf{x}) + \mathcal{L}_{ECL}^{v,\tau}(\mathbf{c}). \quad (12)$$

E. Training and Inference

The model is optimized by minimizing the following loss:

$$\mathcal{L}(\Theta) = \mathcal{L}_{CE} + \gamma_1 \mathcal{L}_{ECL} + \gamma_2 \mathcal{L}_{SPA}, \quad (13)$$

where $\mathcal{L}_{CE}$ denotes the binary cross-entropy loss between the predictions of all modalities from ComplEx [27] and the labels of factual triples. γ_1 and γ_2 are the loss trade-off parameters. In the inference stage, after obtaining plausibility scores from the structural, visual, and textual representations, the boosting fusion strategy [5], [17] is used to obtain the final plausibility scores for performance evaluation.

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

Dataset: Following previous methods, the effectiveness of our framework is evaluated on five benchmark datasets: FB15K-237 [49], WN18RR [30], YAGO15K [50], MKG-W [51], and MKG-Y [51]. The statistical information of these datasets has been shown in Table I. After training, we leverage five metrics,

TABLE I
CHARACTERISTICS STATISTICS OF DATASETS FOR MKGC

Dataset	#Relation	#Entity	#Train	#Valid	#Test
FB15K-237 [49]	237	14,541	272,115	17,535	20,466
WN18RR [30]	11	40,943	86,835	3,034	3,134
YAGO15K [50]	32	15,404	86,020	12,289	24,577
MKG-W [51]	169	15,000	34,196	4,276	4,274
MKG-Y [51]	28	15,000	21,310	2,665	2,663

consisting of Hits@ k ($k \in \{1, 3, 10\}$), Mean Reciprocal Rank (MRR), and Mean Rank (MR), to conduct comparisons.

Implementation Details: For FB15K-237, WN18RR, MKG-W, and MKG-Y, visual and textual embeddings are extracted using the CLIP-ViT-L/14 [42] model. Due to the unavailability of images for YAGO15K [17], the pre-extracted visual embeddings from MMKG [56] are directly leveraged, and BERT [57] is used to extract its textual embeddings. Following [5], [17], the initial embedding dimension is set to 4,000. Based on sensitivity analysis, we determine 5,000 as the optimal default dimension that achieves a trade-off between performance and model complexity. Due to the quality diversity of MKGs, grid search is employed to determine the optimal values of γ_1 and γ_2 from $\{10^{-i} | i = 0, \dots, 4\}$. In the end, (γ_1, γ_2) are fixed at $(10^{-2}, 10^{-3})$ for FB15K-237 and MKG-Y, $(10^0, 10^{-1})$ for WN18RR and MKG-W, $(10^0, 10^{-2})$ for YAGO15K. Additionally, ϑ_p and τ are set to 0.5 and 0.07, respectively. When fitting the GMM, we set the number of mixture components $K = 2$ and the maximum number of EM iterations $I = 100$. This simplifies the computational complexity of GMM optimization from $\mathcal{O}(I \times K \times d_l \times |\mathcal{E}|)$ to a linear complexity $\mathcal{O}(|\mathcal{E}|)$ because I , K , and the scalar loss dimension $d_l = 1$ are much smaller constants than $|\mathcal{E}|$.

B. Main Results and Analysis

The experimental results on the FB15K-237 and WN18RR datasets are summarized in Table II. It is not hard to observe that multimodal methods generally outperform unimodal ones, as visual and textual modalities provide additional informative cues. For multimodal methods, the CL-based methods such as IMF [16] and HKA [17], generally outperform non-contrastive ones. This indirectly indicates the necessity of capturing cross-modal consensus in multimodal entity representations. Importantly, the proposed framework consistently maintains optimal or near-optimal MKGC performance across all metrics on both datasets. On WN18RR, compared with the strongest baseline, HKA [17], SECL demonstrates substantial improvements with Hits@1 increasing by 6.2% and MRR by 5.1%. Meanwhile, impressive improvements can also be observed on FB15K-237. SECL substantially surpasses HKA across all metrics, yielding p -values in paired t -tests of (0.021, 0.032, 0.009, 0.013) for (Hits@1, Hits@3, Hits@10, MRR), respectively, which firmly rejects the null hypothesis at the $\alpha = 0.05$ level.

Moreover, our SECL is compared with state-of-the-art methods on YAGO15K, and the experimental results are presented in Table III. Although images are not provided for our method to extract CLIP features, SECL achieves optimal results across

TABLE II

COMPARISONS ON FB15K-237 AND WN18RR. BEST RECORDS ARE HIGHLIGHTED IN **BOLD** AND SECOND-BEST ONES ARE UNDERLINED. THE RESULTS OF † ARE OBTAINED BY OUR REPRODUCTION. THE MEAN AND STANDARD DEVIATION ARE OBTAINED OVER THREE RUNS.

Method	Venue	FB15K-237					WN18RR				
		Hits@1	Hits@3	Hits@10	MR	MRR	Hits@1	Hits@3	Hits@10	MR	MRR
Unimodal Methods											
TransE [26]	NIPS-2013	0.198	0.376	0.441	323	0.294	0.043	0.441	0.532	2300	0.226
ComplEx [27]	ICML-2016	0.194	0.297	0.450	546	0.247	0.409	0.469	0.530	7882	0.440
ConvE [30]	AAAI-2018	0.225	0.341	0.497	245	0.325	0.419	0.470	0.531	4464	0.430
RotatE [52]	ICLR-2019	0.241	0.375	0.533	177	0.338	0.428	0.492	0.571	3340	0.476
B-CP [53]	TKDE-2023	0.210	0.324	0.483	-	0.295	0.450	0.490	0.530	-	0.480
TGformer [9]	TKDE-2025	0.279	0.410	0.557	-	0.372	<u>0.455</u>	0.509	0.566	-	0.493
Multimodal Methods											
IKLR [39]	IJCAI-2017	0.194	0.284	0.458	298	0.309	0.393	0.436	0.491	2760	-
RSME [40]	MM-2021	0.242	0.344	0.467	417	-	0.423	0.486	0.531	1571	-
MKGformer [11]	SIGIR-2022	0.256	0.367	0.504	221	-	0.438	0.520	0.615	362	-
†MoSE [5]	EMNLP-2022	0.279	0.409	0.562	117	0.373	0.416	<u>0.560</u>	<u>0.678</u>	105	0.509
IMF [16]	WWW-2023	0.287	-	0.593	134	<u>0.389</u>	0.440	<u>0.487</u>	0.542	2803	-
SGMPT [41]	MM-2024	0.252	0.370	0.510	238	-	-	-	-	-	-
LAFA [12]	AAAI-2024	0.262	0.386	0.540	157	-	-	-	-	-	-
CMR [54]	SIGIR-2024	0.263	0.395	0.543	123	-	-	-	-	-	-
MoCi [14]	MM-2024	0.263	0.395	0.543	123	-	-	-	-	-	-
†HKA [17]	TOMM-2024	<u>0.290</u>	<u>0.425</u>	0.577	<u>110</u>	0.386	0.449	0.559	0.675	<u>97</u>	<u>0.526</u>
†MoMoK [15]	ICLR-2025	0.267	0.397	0.551	160	0.361	0.428	0.484	0.541	3766	0.468
†MyGO [13]	AAAI-2025	0.267	0.396	0.544	139	0.359	0.436	0.491	0.543	1844	0.475
SECL	Ours	0.295 ±0.001	0.428 ±0.001	<u>0.583</u> ±0.001	103 ±1	0.390 ±0.001	0.477 ±0.007	0.583 ±0.007	0.704 ±0.004	87 ±4	0.553 ±0.006

TABLE III

COMPARISONS WITH STATE-OF-THE-ART METHODS ON YAGO15K

Method	Hits@1	Hits@3	Hits@10	MR	MRR
TransE [26]	0.228	-	0.468	614	0.164
DistMult [55]	0.215	-	0.438	578	0.365
ComplEx [27]	0.291	0.360	0.451	692	-
ConvE [30]	0.336	0.434	0.518	536	0.409
RotatE [52]	0.289	0.391	0.479	540	<u>0.455</u>
IKLR [39]	0.162	0.247	0.346	548	0.139
RSME [40]	0.233	0.319	0.414	364	-
MoSE [5]	0.361	0.463	0.565	213	0.429
IMF [16]	0.276	-	0.490	488	0.345
MoCi [14]	0.369	0.462	0.545	-	0.430
HKA [17]	<u>0.381</u>	<u>0.485</u>	<u>0.590</u>	189	0.451
SECL	0.384 ±0.001	0.495 ±0.001	0.604 ±0.001	189 ±1	0.461 ±0.000

TABLE IV

COMPARISONS ON SMALL-SCALE DATASETS

Method	MKG-W			MKG-Y		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
TransE [26]	0.211	0.442	0.292	0.235	0.434	0.307
DistMult [55]	0.159	0.309	0.210	0.193	0.360	0.250
ComplEx [27]	0.191	0.367	0.249	0.223	0.409	0.287
RotatE [52]	0.268	0.467	0.337	0.291	0.453	0.350
IKLR [39]	0.261	0.441	0.324	0.304	0.383	0.332
RSME [40]	0.234	0.404	0.292	0.318	0.391	0.344
MoSE [5]	0.278	0.411	0.333	0.336	0.408	0.363
MMRNS [51]	0.286	0.475	0.350	0.305	0.455	0.359
IMF [16]	0.288	0.454	0.345	0.330	0.406	0.358
QEB [58]	0.255	0.453	0.324	0.295	0.423	0.344
MoCi [14]	0.307	0.459	0.358	<u>0.356</u>	0.450	0.389
HKA [17]	<u>0.325</u>	<u>0.577</u>	<u>0.413</u>	0.334	<u>0.553</u>	<u>0.411</u>
MoMoK [15]	0.304	0.461	0.359	0.351	0.432	0.379
MyGO [13]	0.298	0.478	0.361	0.350	0.442	0.384
SECL	0.349 ±0.003	0.588 ±0.002	0.432 ±0.001	0.370 ±0.006	0.572 ±0.001	0.442 ±0.005

all metrics. Compared with the strongest baseline, HKA [17], the proposed framework substantially surpasses it, backed by p -values of (0.023, 0.007, 0.001, 0.002) for (Hits@1, Hits@3, Hits@10, MRR) that firmly reject the null hypothesis.

To demonstrate the generalization of SECL on small-scale datasets, it is compared with state-of-the-art methods on MKG-W and MKG-Y, as shown in Table IV. Obviously, SECL still yields substantially superior performance over all state-of-the-art approaches across all evaluation metrics. On the MKG-W dataset, it obtains relative gains of 7.4% in Hits@1 and 4.6% in MRR, and it also achieves relative gains of 3.9% and 7.5% in the same metrics on MKG-Y. The high performance of our framework on both large-scale and small-scale datasets shows its excellent generalization capabilities.

Model complexity is important in the deployment of models in real-world scenarios. As shown in Table V, SECL achieves state-of-the-art performance without excessive parameter size,

memory footprint, and wall-clock time. Evidently, our framework does not introduce unacceptable computational overhead, achieving an excellent trade-off between efficiency and performance.

C. Cross-Modality Alignment Analysis

This section further conducts experiments to verify whether SECL is faced with modality degradation on different datasets. Practically, a certain modality pair is excluded from the full objective in training to verify whether this results in performance improvements. The experimental results have been provided in Table VI. It is not hard to observe that each pair of modalities makes a positive contribution to our SECL. Across all metrics, jointly aligning all pairwise modalities in $\mathcal{P}$ achieves optimal MKGC performance. These results substantiate that SECL can

TABLE V

COMPARISON OF PARAMETER SIZE (M), MEMORY USAGE (GB), PER-EPOCH TIME (S), AND TOTAL TIME (H) ACROSS DIFFERENT BASELINES. RESULTS ARE OBTAINED ON A SINGLE RTX 3090 GPU. “-” DENOTES THAT THE DATA IS UNAVAILABLE DUE TO OUT-OF-MEMORY.

Method	FB15K-237				WN18RR			
	#Para.	Mem.	Epoch	Total	#Para.	Mem.	Epoch	Total
MKGformer	237.63	OOM	-	-	257.45	OOM	-	-
MoSE	70.92	3.21	50.51	0.72	171.11	8.28	41.70	0.60
HKA	70.43	4.44	69.50	1.11	171.41	11.76	52.44	0.58
MoMoK	63.77	3.84	20.42	11.34	82.24	6.20	24.98	13.88
MyGO	46.84	12.53	93.79	39.08	53.55	21.22	53.14	22.14
SECL	103.32	6.87	115.12	1.22	229.34	18.05	107.09	4.19

capture informative cues from multimodal data and achieve modality evolution instead of degradation.

Furthermore, to intuitively demonstrate the effectiveness of SECL in enhancing the trustworthiness of auxiliary modalities, extensive additional experiments are conducted, and the results are provided in Table VII. As we can notice from the “Base” model, there is an inherent trustworthiness gap across different modalities, as reflected by their respective performance. When these native multimodal representations are aligned (w/ Native), performance degradation is observed across different modalities. This effect is more evident on WN18RR, where the trustworthiness gap between modalities is significantly larger than that on FB15K-237. During alignment, since semantically irrelevant signals propagate between different modalities, trustworthy modalities are pulled down by untrustworthy ones, thus driving the representations toward mediocrity and resulting in a deterioration of overall performance. In contrast, our SECL consistently improves the performance of all modalities across all metrics, which confirms its effectiveness in improving the trustworthiness of multimodal representations.

D. Ablation Study

Study on SMRH: As shown in Table VIII, SMRH markedly contributes to the performance of our framework on different MKGC datasets. On WN18RR, when SMRH is removed from the full framework, there is a significant degradation of 7.5% in Hits@1 and 6.1% in MRR. This can be attributed to the fact that numerous entities in the WN18RR dataset fall short of images, and their textual descriptions are coarse-grained. If CL is applied blindly to align these modalities, the originally high-quality representations will deteriorate, thereby impairing the overall MKGC performance. A similar performance drop can also be observed on FB15K-237 and YAGO15K. SMRH effectively tackles this problem with the help of the structural context of entities. To provide a more intuitive understanding of SMRH, the statistics of harmonizer coefficients are depicted in Fig. 10. Obviously, a large number of coefficients are around 0, suggesting that lots of semantically irrelevant features have been suppressed.

As a plug-and-play module, SMRH exhibits high flexibility and compatibility with other methods. To verify this statement, additional experiments are conducted by integrating it into two state-of-the-art methods, i.e., MoSE [5] and HKA [17]. The

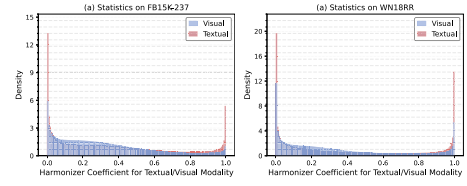


Fig. 10. Statistics of the harmonizer coefficients.

experimental results are summarized in Table IX. It is obvious that incorporating SMRH consistently boosts the performance of both methods across all metrics. These results demonstrate the effectiveness and generalizability of SMRH, indicating that our method can be seamlessly incorporated into other methods to enhance the entity representation quality.

Study on ECL: To demonstrate the necessity of ECL, ablation experiments are conducted on diverse datasets. As we can see from Table VIII, it consistently harms the performance of our model when ECL is excluded from the training procedure. The drop in performance underscores the importance of capturing cross-modal consensus in multimodal representations. An interesting observation is that, when ECL is eliminated from the full framework, the performance of our model is comparable to that of the baseline (w/o all) on YAGO15K. This confirms our hypothesis in Section IV-C: It is insufficiently reliable to suppress semantically irrelevant features for debiasing only by SMRH. With the powerful supervisory signals from ECL, SMRH can overcome this problem and achieve improved performance. In addition, ECL is compared to InfoNCE in Table X. Obviously, traditional InfoNCE treats each entity equally, overlooking the different optimization difficulties among entities and thus still resulting in modality degradation. In contrast, our method can tackle it by balancing the optimization of easy and hard entities (see Table VI).

E. In-Depth Study Under Challenging Scenarios

In this subsection, additional noise is deliberately introduced to conduct an in-depth study on the proposed framework under more challenging scenarios. Specifically, we conduct two types of experiments, including *modality missing* as well as *semantic corruption*. For the former, a subset of entities is first randomly selected, and their visual and textual embeddings are replaced with vectors sampled from a normal distribution. For the latter, we randomly select entities and replace their visual and textual embeddings with those of other randomly selected entities. The results are illustrated in Fig. 11, where the selection percentage is varied from 10% to 50% to impose increasing noise. Finally, we study the performance under a severe scenario of relation incompleteness.

Under the modality missing scenario in Fig. 11(upper), as the missing percentage increases, the performance of different methods tends to converge. This is expected since the absence of visual and textual representations leaves MKGs dominated by structural information, thus narrowing the performance gap. Meanwhile, this trend highlights the superior efficiency of our method in capturing valuable information from multimodal data.

TABLE VI

IMPACT OF CROSS-MODAL ALIGNMENT ON PERFORMANCE. $\checkmark$ AND $\times$ REPRESENT WHETHER TO ALIGN THE CORRESPONDING PAIR OF MODALITIES.

$S \leftrightarrow T$	$S \leftrightarrow V$	$T \leftrightarrow V$	FB15K-237				WN18RR			
			Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR
$\checkmark$	$\checkmark$	$\checkmark$	0.295 ± 0.001	0.428 ± 0.001	0.583 ± 0.001	0.390 ± 0.001	0.477 ± 0.007	0.583 ± 0.007	0.704 ± 0.004	0.553 ± 0.006
$\checkmark$	$\checkmark$	$\times$	0.292 ± 0.001	0.426 ± 0.000	0.581 ± 0.001	0.389 ± 0.000	0.465 ± 0.006	0.577 ± 0.001	0.698 ± 0.005	0.544 ± 0.005
$\checkmark$	$\times$	$\checkmark$	0.293 ± 0.000	0.425 ± 0.002	0.582 ± 0.001	0.388 ± 0.001	0.467 ± 0.002	0.580 ± 0.004	0.699 ± 0.001	0.546 ± 0.002
$\times$	$\checkmark$	$\checkmark$	0.292 ± 0.001	0.427 ± 0.001	0.581 ± 0.001	0.388 ± 0.000	0.455 ± 0.011	0.574 ± 0.005	0.695 ± 0.003	0.537 ± 0.007

TABLE VII

EVALUATION OF VISUAL AND TEXTUAL MODALITIES. “BASE” AND “NATIVE” DENOTE MODELS WITH OR WITHOUT INFO NCE LOSS.

Modality	Method	FB15K-237				WN18RR			
		Hits@1	Hits@3	Hits@10	MRR	Hits@1	Hits@3	Hits@10	MRR
V	Base	0.233 ± 0.001	0.348 ± 0.001	0.490 ± 0.001	0.319 ± 0.001	0.366 ± 0.002	0.391 ± 0.003	0.421 ± 0.001	0.385 ± 0.002
	w/ Native	0.234 ± 0.002	0.347 ± 0.003	0.488 ± 0.003	0.319 ± 0.002	0.313 ± 0.011	0.374 ± 0.008	0.432 ± 0.005	0.356 ± 0.008
	w/ SECL	0.253 ± 0.002	0.376 ± 0.001	0.527 ± 0.002	0.344 ± 0.001	0.385 ± 0.001	0.421 ± 0.001	0.466 ± 0.004	0.413 ± 0.001
T	Base	0.265 ± 0.002	0.383 ± 0.000	0.521 ± 0.000	0.351 ± 0.001	0.182 ± 0.002	0.336 ± 0.003	0.513 ± 0.001	0.293 ± 0.002
	w/ Native	0.261 ± 0.002	0.380 ± 0.001	0.516 ± 0.001	0.347 ± 0.001	0.174 ± 0.002	0.329 ± 0.002	0.506 ± 0.002	0.285 ± 0.001
	w/ SECL	0.283 ± 0.002	0.412 ± 0.001	0.566 ± 0.001	0.377 ± 0.001	0.452 ± 0.009	0.575 ± 0.011	0.696 ± 0.006	0.536 ± 0.008

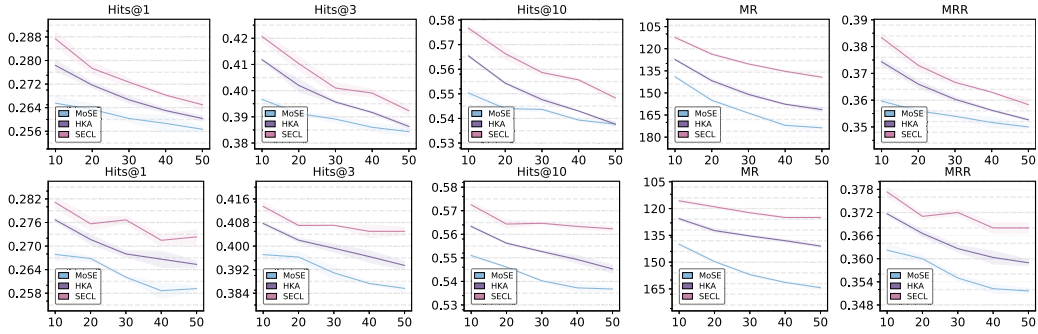
Fig. 11. Results under (upper) modality missing and (lower) semantic corruption scenarios. The x - and y -axis are the percentage and performance, respectively.

TABLE VIII

ABLATION STUDY ON DIFFERENT DATASETS

Dataset	Method	Hits@1	Hits@3	Hits@10	MRR
FB15K-237	SECL	0.295 ± 0.001	0.428 ± 0.001	0.583 ± 0.001	0.390 ± 0.001
	w/o SMRH	0.292 ± 0.000	0.425 ± 0.001	0.578 ± 0.001	0.387 ± 0.000
	w/o ECL	0.290 ± 0.001	0.423 ± 0.000	0.581 ± 0.001	0.386 ± 0.001
	w/o all	0.289 ± 0.000	0.422 ± 0.001	0.576 ± 0.001	0.385 ± 0.000
WN18RR	SECL	0.477 ± 0.007	0.583 ± 0.007	0.704 ± 0.004	0.553 ± 0.006
	w/o SMRH	0.441 ± 0.002	0.553 ± 0.005	0.670 ± 0.008	0.519 ± 0.003
	w/o ECL	0.444 ± 0.007	0.565 ± 0.003	0.685 ± 0.005	0.526 ± 0.006
	w/o all	0.411 ± 0.006	0.555 ± 0.003	0.674 ± 0.002	0.508 ± 0.007
YAGO15K	SECL	0.384 ± 0.001	0.495 ± 0.001	0.604 ± 0.001	0.461 ± 0.000
	w/o SMRH	0.377 ± 0.000	0.484 ± 0.001	0.591 ± 0.002	0.452 ± 0.001
	w/o ECL	0.367 ± 0.008	0.481 ± 0.001	0.589 ± 0.001	0.444 ± 0.005
	w/o all	0.372 ± 0.005	0.478 ± 0.002	0.585 ± 0.001	0.446 ± 0.003

TABLE IX

PLUG-AND-PLAY VALIDATION OF SMRH

Dataset	Method	Hits@1	Hits@3	Hits@10	MRR
FB15K-237	MoSE [5]	0.279 ± 0.002	0.409 ± 0.003	0.562 ± 0.002	0.373 ± 0.002
	w/ SMRH	0.284 ± 0.003	0.417 ± 0.001	0.571 ± 0.001	0.380 ± 0.001
WN18RR	HKA [17]	0.290 ± 0.001	0.425 ± 0.001	0.577 ± 0.001	0.386 ± 0.001
	w/ SMRH	0.293 ± 0.001	0.425 ± 0.001	0.581 ± 0.001	0.388 ± 0.001
WN18RR	MoSE [5]	0.416 ± 0.004	0.560 ± 0.001	0.678 ± 0.001	0.509 ± 0.002
	w/ SMRH	0.458 ± 0.009	0.553 ± 0.008	0.653 ± 0.006	0.525 ± 0.007
WN18RR	HKA [17]	0.449 ± 0.002	0.559 ± 0.002	0.675 ± 0.001	0.526 ± 0.002
	w/ SMRH	0.464 ± 0.004	0.583 ± 0.003	0.697 ± 0.003	0.547 ± 0.004

Under the semantic corruption scenario in Fig. 11(lower), SECL has more stable behavior with a lower degradation rate. Compared to HKA [17], which suffers substantial performance drops (e.g., Hits@1 and MRR dropping by 4.10% and 3.42%), SECL limits these degradations to 3.12% and 2.47%, respectively. These experimental results demonstrate the robustness of the proposed framework.

Fig. 12 depicts the performance when 50% of the relations in the MKG are randomly removed. It is not difficult to notice

that the proposed framework still yields optimal performance. This shows that our framework has strong resilience to structural noise as well.

F. Sensitivity Analysis

Analysis of Balance Term: The hyper-parameters γ in $\mathcal{L}(\Theta)$ have an impact on the performance of SECL. To explore how they correlate with performance, we provide the experimental results under different hyper-parameters according to the grid search in Fig. 13. It can be observed that the best performance

TABLE X
COMPARISON BETWEEN OUR METHOD AND INFO NCE ON WN18RR

Method	Hits@1	Hits@3	Hits@10	MRR
InfoNCE [20]	0.453 $\pm$ 0.002	0.572 $\pm$ 0.001	0.693 $\pm$ 0.004	0.536 $\pm$ 0.002
w/o $\mathcal{T} \leftrightarrow \mathcal{V}$	0.466 $\pm$ 0.010	0.576 $\pm$ 0.004	0.696 $\pm$ 0.005	0.544 $\pm$ 0.007
w/o $\mathcal{S} \leftrightarrow \mathcal{T}$	0.464 $\pm$ 0.000	0.579 $\pm$ 0.000	0.698 $\pm$ 0.007	0.544 $\pm$ 0.001
w/o $\mathcal{S} \leftrightarrow \mathcal{V}$	0.454 $\pm$ 0.014	0.568 $\pm$ 0.005	0.691 $\pm$ 0.009	0.534 $\pm$ 0.010
ECL	0.477$\pm$0.007	0.583$\pm$0.007	0.704$\pm$0.004	0.553$\pm$0.006

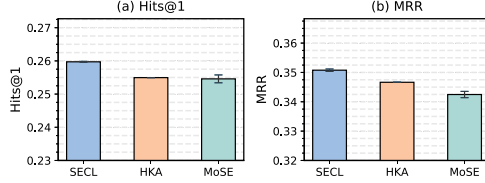


Fig. 12. Evaluation under 50% of the relations within the FB15K-237 dataset.

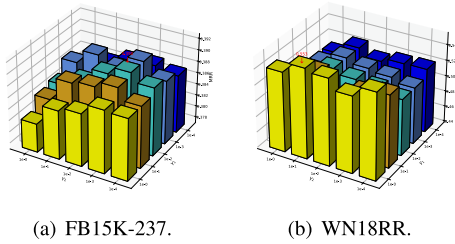


Fig. 13. Influence of γ_1 and γ_2 on model performance.

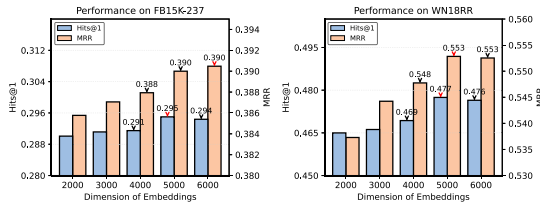


Fig. 14. Influence of embedding dimension. The best are highlighted in red.

is observed at $(10^{-2}, 10^{-3})$ on FB15K-237 and at $(10^0, 10^{-1})$ on WN18RR for (γ_1, γ_2) . The higher magnitudes required for WN18RR compared with FB15K-237 can be attributed to the fact that the visual and textual information associated with the entities in WN18RR is coarser. Therefore, stronger contrastive and sparsity losses are necessary for our framework to model trustworthy multimodal representations.

Analysis of Embedding Dimension: The embedding dimension is an important factor that determines the representational capacity of MKGC models. Therefore, we conduct a number of experiments to find the optimal embedding dimension. The results are illustrated in Fig. 14. As we can see, increasing the embedding dimension demonstrates stronger performance, but with diminishing returns. When the dimension is set to 4,000, consistent with our baseline methods, (e.g., MoSE and HKA), SECL can still achieve optimal or near-optimal performance across all evaluation metrics in comparison to the baselines in

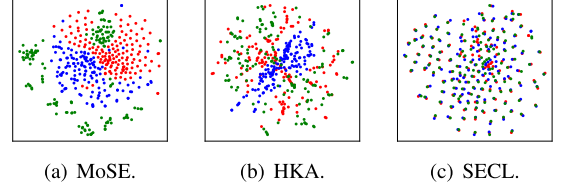


Fig. 15. t-SNE plots of learned representations. Red, blue, and green points indicate structural, visual, and textual representations, respectively.

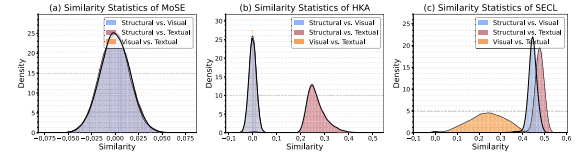


Fig. 16. Statistics of the cosine similarity across different modality pairs.

Table II. The best performance is observed at 5,000 and 6,000. To achieve a trade-off between performance and complexity, we set the default embedding dimension to 5,000.

G. Visualization of Embeddings

To intuitively assess the quality of the embeddings learned by different methods, Fig. 15 shows the t-SNE plots of entity embeddings from structural, visual, and textual modalities on the WN18RR dataset [30]. To promote observation, every 100-th entity is selected for visualization. As observed, MoSE [5] keeps a significant modality discrepancy, as cross-modal consensus is not captured. Although HKA [17] achieves improved performance than MoSE, a notable modality discrepancy still exists between the structural and visual embeddings, indicating insufficient exploration of cross-modal consensus. Conversely, our framework mitigates this problem. The similarity distribution is shown in Fig. 16. Compared to baselines, our method achieves markedly better alignment across all modality pairs.

VI. CONCLUSION

In this paper, we show that the modality degradation could diminish the synergy between contrastive learning and MKGC. To mitigate this issue, this paper introduces a Structure-Guided Elastic Consensus Learning (SECL) framework, which purifies untrustworthy multimodal entity representations into trustworthy ones, such that preventing high-performing representations from being corrupted during contrastive learning. Specifically, this is achieved by selectively suppressing semantically irrelevant features in multimodal representations at an entity level under the explicit supervision from elastic contrastive learning. Extensive experiments on benchmark datasets show that SECL significantly improves the trustworthiness of auxiliary modalities, promote modality evolution rather than degradation, and yield consistent improvements over state-of-the-art baselines.

Limitation and Future Work: Although our method has shown superior performance against state-of-the-art methods, it still has a limitation. In particular, large multimodal models (LMMs)

(e.g., LLaVA [59]) have emerged as a central focus in multimodal learning over the past years. Nevertheless, the rich cross-modal knowledge encapsulated in these models remains untapped in this study. Exploring how to use the sophisticated reasoning capability of LMMs to facilitate MKGC presents a promising research direction.

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